

```
!pip install transformers==4.40.2
!pip install torch --quiet
!pip install sentencepiece --quiet
```

```
Collecting transformers==4.40.2
  Downloading transformers-4.40.2-py3-none-any.whl.metadata (137 kB)
    138.0/138.0 kB 8.6 MB/s eta 0:00:00
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from transformers==4.40.2) (3.25.2)
Collecting huggingface-hub<1.0,>=0.19.3 (from transformers==4.40.2)
  Downloading huggingface_hub-0.36.2-py3-none-any.whl.metadata (15 kB)
Requirement already satisfied: numpy>=1.17 in /usr/local/lib/python3.12/dist-packages (from transformers==4.40.2) (2.0.2)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from transformers==4.40.2) (26.0)
Requirement already satisfied: pyyaml>=5.1 in /usr/local/lib/python3.12/dist-packages (from transformers==4.40.2) (6.0.3)
Requirement already satisfied: regex!=2019.12.17 in /usr/local/lib/python3.12/dist-packages (from transformers==4.40.2) (2025.11.3)
Requirement already satisfied: requests in /usr/local/lib/python3.12/dist-packages (from transformers==4.40.2) (2.32.4)
Collecting tokenizers<0.20,>=0.19 (from transformers==4.40.2)
  Downloading tokenizers-0.19.1-cp312-cp312-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (6.7 kB)
Requirement already satisfied: safetensors>=0.4.1 in /usr/local/lib/python3.12/dist-packages (from transformers==4.40.2) (0.7.0)
Requirement already satisfied: tqdm>=4.27 in /usr/local/lib/python3.12/dist-packages (from transformers==4.40.2) (4.67.3)
Requirement already satisfied: fsspec>=2023.5.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<1.0,>=0.19.3->transformers==4.40.2) (2025.3.0)
Requirement already satisfied: hf-xet<2.0.0,>=1.1.3 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<1.0,>=0.19.3->transformers==4.40.2) (1.4.2)
Requirement already satisfied: typing-extensions>=3.7.4.3 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<1.0,>=0.19.3->transformers==4.40.2) (4.15.0)
Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests->transformers==4.40.2) (3.4.6)
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages (from requests->transformers==4.40.2) (3.11)
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests->transformers==4.40.2) (2.5.0)
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-packages (from requests->transformers==4.40.2) (2026.2.25)
Downloading transformers-4.40.2-py3-none-any.whl (9.0 MB)
    9.0/9.0 MB 75.0 MB/s eta 0:00:00
Downloading huggingface_hub-0.36.2-py3-none-any.whl (566 kB)
    566.4/566.4 kB 36.6 MB/s eta 0:00:00
Downloading tokenizers-0.19.1-cp312-cp312-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (3.6 MB)
    3.6/3.6 MB 74.5 MB/s eta 0:00:00
Installing collected packages: huggingface-hub, tokenizers, transformers
  Attempting uninstall: huggingface-hub
    Found existing installation: huggingface_hub 1.7.1
    Uninstalling huggingface_hub-1.7.1:
      Successfully uninstalled huggingface_hub-1.7.1
  Attempting uninstall: tokenizers
    Found existing installation: tokenizers 0.22.2
    Uninstalling tokenizers-0.22.2:
      Successfully uninstalled tokenizers-0.22.2
  Attempting uninstall: transformers
    Found existing installation: transformers 5.0.0
    Uninstalling transformers-5.0.0:
      Successfully uninstalled transformers-5.0.0
ERROR: pip's dependency resolver does not currently take into account all the packages that are installed. This behaviour is the source of the following dependency conf
sentence-transformers 5.3.0 requires transformers<6.0.0,>=4.41.0, but you have transformers 4.40.2 which is incompatible.
Successfully installed huggingface-hub-0.36.2 tokenizers-0.19.1 transformers-4.40.2
```

```
from transformers import pipeline
```

```
summarizer = pipeline("summarization")
```

```
No model was supplied, defaulted to sshleifer/distilbart-cnn-12-6 and revision a4f8f3e (https://huggingface.co/sshleifer/distilbart-cnn-12-6).
Using a pipeline without specifying a model name and revision in production is not recommended.
/usr/local/lib/python3.12/dist-packages/huggingface_hub/file_download.py:949: FutureWarning: `resume_download` is deprecated and will be removed in version 1.0.0. Downl
warnings.warn(
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:94: UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (https://huggingface.co/settings/tokens), set it as secret in your Google Colab and resta
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or datasets.
warnings.warn(
config.json:      1.80k/? [00:00<00:00, 95.7kB/s]

pytorch_model.bin: 100%                               1.22G/1.22G [00:09<00:00, 174MB/s]

tokenizer_config.json: 100%                             26.0/26.0 [00:00<00:00, 2.54kB/s]

vocab.json:      899k/? [00:00<00:00, 23.5MB/s]

merges.txt:      456k/? [00:00<00:00, 17.0MB/s]
```

```
text = """"Artificial Intelligence is transforming the world by enabling machines to perform tasks that usually require human intelligence such as learning, reasoning,

summary = summarizer(text, max_length=50, min_length=20, do_sample=False)

print(summary[0]['summary_text'])
```

Your max_length is set to 50, but your input_length is only 29. Since this is a summarization task, where outputs shorter than the input are typically wanted, you might Artificial Intelligence is transforming the world by enabling machines to perform tasks that usually require human intelligence such as learning, reasoning, and decisi

```
!pip install transformers==4.40.2
!pip install torch --quiet
!pip install sentencepiece --quiet

from transformers import pipeline

summarizer = pipeline("summarization", device=-1)

text = """"Artificial Intelligence is transforming industries by enabling machines to perform tasks that typically require human intelligence. It is widely used in heal

summary = summarizer(text[:500], max_length=50, min_length=20, do_sample=False)

print(summary[0]['summary_text'])
```

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Requirement already satisfied: transformers==4.40.2 in /usr/local/lib/python3.12/dist-packages (4.40.2)
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from transformers==4.40.2) (3.25.2)
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Requirement already satisfied: numpy>=1.17 in /usr/local/lib/python3.12/dist-packages (from transformers==4.40.2) (2.0.2)
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Requirement already satisfied: tokenizers<0.20,>=0.19 in /usr/local/lib/python3.12/dist-packages (from transformers==4.40.2) (0.19.1)
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Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages (from requests->transformers==4.40.2) (3.11)
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Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-packages (from requests->transformers==4.40.2) (2026.2.25)
No model was supplied, defaulted to sshleifer/distilbart-cnn-12-6 and revision a4f8f3e (https://huggingface.co/sshleifer/distilbart-cnn-12-6).
Using a pipeline without specifying a model name and revision in production is not recommended.
/usr/local/lib/python3.12/dist-packages/huggingface_hub/file_download.py:949: FutureWarning: `resume_download` is deprecated and will be removed in version 1.0.0. Downl
warnings.warn(
Your max_length is set to 50, but your input_length is only 31. Since this is a summarization task, where outputs shorter than the input are typically wanted, you might
Artificial Intelligence is transforming industries by enabling machines to perform tasks that typically require human intelligence . It is widely used in healthcare, f
```

```
text1 = """Paste a news article (150-300 words)"""
text2 = """Paste a technical paragraph"""
text3 = """Paste a story"""
text4 = """Paste a conversation"""
text5 = """Paste a blog or general content"""
```

```
texts = [text1, text2, text3, text4, text5]
```

```
for i, txt in enumerate(texts):
    print(f"\n===== TEXT {i+1} =====\n")
    print(txt)

    summary = summarizer(txt[:800], max_length=100, min_length=30, do_sample=False)

    print(f"\n----- SUMMARY {i+1} ----- \n")
    print(summary[0]['summary_text'])
```

Your max_length is set to 100, but your input_length is only 12. Since this is a summarization task, where outputs shorter than the input are typically wanted, you might

===== TEXT 1 =====

Paste a news article (150-300 words)

Your max_length is set to 100, but your input_length is only 6. Since this is a summarization task, where outputs shorter than the input are typically wanted, you might

----- SUMMARY 1 -----

Paste a news article (150-300 words) Share a photo of yourself with a caption . Use this article to help us understand today's featured news stories . Use the weekly N

===== TEXT 2 =====

Paste a technical paragraph

Your max_length is set to 100, but your input_length is only 5. Since this is a summarization task, where outputs shorter than the input are typically wanted, you might

----- SUMMARY 2 -----

Paste a technical paragraph here to help you understand how to use the Internet . Use the weekly Newsquiz to test your knowledge of stories you saw on CNN iReport . Vi

===== TEXT 3 =====

Paste a story

Your max_length is set to 100, but your input_length is only 5. Since this is a summarization task, where outputs shorter than the input are typically wanted, you might

----- SUMMARY 3 -----

CNN.com will feature iReporter photos in a weekly Travel Snapshots gallery . Please submit a photo of your favorite travel destinations for next week . Share your phot

===== TEXT 4 =====

Paste a conversation

Your max_length is set to 100, but your input_length is only 8. Since this is a summarization task, where outputs shorter than the input are typically wanted, you might

----- SUMMARY 4 -----

CNN.com will feature iReporter photos in a weekly Travel Snapshots gallery . Please submit a photo of your favorite travel destinations for next week's featured travel

===== TEXT 5 =====

Paste a blog or general content

----- SUMMARY 5 -----

Paste a blog or general content into your blog post . Use this image to help us understand today's featured content . Use the weekly Newsquiz to test your knowledge of

```
text = text1
```

```
print("\nSHORT SUMMARY:\n")
print(summarizer(text[:800], max_length=50, min_length=20, do_sample=False)[0]['summary_text'])
```

```
print("\nMEDIUM SUMMARY:\n")
print(summarizer(text[:800], max_length=100, min_length=40, do_sample=False)[0]['summary_text'])
```

```
print("\nLONG SUMMARY:\n")
print(summarizer(text[:800], max_length=150, min_length=60, do_sample=False)[0]['summary_text'])
```

Your max_length is set to 50, but your input_length is only 12. Since this is a summarization task, where outputs shorter than the input are typically wanted, you might

SHORT SUMMARY:

Your max_length is set to 100, but your input_length is only 12. Since this is a summarization task, where outputs shorter than the input are typically wanted, you might
Paste a news article (150-300 words) Share a photo of yourself with a caption . Use this article to help us understand today's featured news stories . Use the weekly N

MEDIUM SUMMARY:

Your max_length is set to 150, but your input_length is only 12. Since this is a summarization task, where outputs shorter than the input are typically wanted, you might
Paste a news article (150-300 words) Share a photo of yourself with a caption . Use this article to help us understand today's featured news stories . Use the weekly N

LONG SUMMARY:

Paste a news article (150-300 words) Share a photo of yourself with a caption . Use this article to help us understand today's featured news stories . Use the weekly N

```
long_text = """Paste a very long paragraph (400+ words)"""
```

```
summary = summarizer(long_text[:1000], max_length=150, min_length=50, do_sample=False)
```

```
print(summary[0]['summary_text'])
```

Your max_length is set to 150, but your input_length is only 12. Since this is a summarization task, where outputs shorter than the input are typically wanted, you might
Paste a very long paragraph (400+ words) into a long sentence . Paste a long paragraph into a large paragraph . Copy the entire text and paste it into a new paragraph.

